M&A Risk Analyzer

A Quantitative framework for screening acquisition targets

Single Target Assessment, Merger Analysis, and Textual Sentiment Analysis

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Business model & solution

PROBLEM

Traditional M&A due diligence is slow, expensive, and relies heavily on manual financial analysis.

→

SOLUTION

M&A Risk Analyzer — a web-based platform that instantly assesses bankruptcy risk, earnings manipulation, and merger synergy.

TARGET USERS

- M&A professionals
- Private equity investors
- Corporate development teams
- Financial analysts

CORE VALUE

- ⚡ One-click full analysis
- 🤖 Industry-specific ML models
- 📄 NLP on 10-K filings

What the model is doing

Six models running simultaneously — each tackling a different dimension of financial risk.

01 Distress Scoring

7 industry-specific XGBoost models predict bankruptcy risk.

02 Manipulation Detection

Beneish M-Score flags potential earnings manipulation.

03 Bankruptcy Probability

Logistic Regression estimates probability of financial distress.

04 Merger Synergy Analysis

Evaluates risk and strategic fit between acquirer and target.

05 Textual Sentiment NLP

DistilBERT + Loughran-McDonald dictionary analyzes 10-K filings.

06 Valuation

Estimates intrinsic and fair value of any company.

Models that Powered the App

Six models running simultaneously — each tackling a different dimension of financial risk.

01 Beneish M-Score

- -4.84 + 0.92·DSRI + 0.528·GMI + 0.404·AQI + 0.892·SGI + 0.115·DEPI - 0.172·SGAI + 4.679·ACCR + 0.327 ·LEVI.
- M > -1.78 = Likely Manipulator.

02 Logistic Regression

- P(Bankruptcy)=1/(1+e^-(β0+β1X1+...+βnXn)).
- Higher probability = higher bankruptcy risk.

03 XGBooster Distress Score (ISDS)

- XGBoost Distress: P(Distress) = 1/(1+e^-raw_score).
- Outputs Safe / Grey / Distress zones.

04 CAPM

- Beta: β = Cov(Rstock,Rmarket)/Var(Rmarket
- CAPM: Ke = Rf + β×(Rm-Rf).

05 Valuation

- Graham Fair Value = √(22.5 ×EPS×BVPS).
- DCF = Sum of discounted FCF + terminal value.

06 Textual Sentiment Analysis

- P(Positive Sentiment)=1/(1+e^-(β0+β1X1+β2X2+...))
- L&M + DistilBERT features
- Flesch-Kincaid
- Gunning Fog
- ARI
- Output: Sentiment score + phrases + readability grade.

Training the data

Six models running simultaneously — each tackling a different dimension of financial risk.

01 Data Source

Compustat database (WRDS) with over 70 years of historical financial data

02 Industry-Specific Training

Separate XGBoost models trained for each of the 7 industries using sector-specific financial ratios

03 Target Variable

Binary distress label (costat = 'I' = inactive/distressed vs 'A' = active)

04 Model Used

XGBoost Classifier with industry-calibrated probability thresholds for Safe / Grey / Distress zones

05 Additional Models

- Beneish M-Score (rule-based 8-ratio model)
- Logistic Regression for bankruptcy probability

06 Textual Analysis

Trained on Loughran & McDonald sentiment dictionary + DistilBERT embeddings

XGBoost Industry Distress Models – Performance Summary

7 Industry-Specific Models | Trained on Real Compustat Data | ISDS XGBoost Real Data Report

Industry	AUC	Accuracy	Safe Zone	Grey Zone	Distress Zone
Healthcare	0.695	0.641	$p < 0.471$	0.471 – 0.543	$p > 0.543$
Technology	0.682	0.658	$p < 0.499$	0.499 – 0.555	$p > 0.555$
Manufacturing	0.696	0.655	$p < 0.507$	0.507 – 0.545	$p > 0.545$
Energy	0.664	0.635	$p < 0.442$	0.442 – 0.603	$p > 0.603$
Construction & Real Estate	0.749	0.689	$p < 0.471$	0.471 – 0.549	$p > 0.549$
Airlines	0.869	0.747	$p < 0.462$	0.462 – 0.618	$p > 0.618$
Agriculture & Food	0.772	0.696	$p < 0.481$	0.481 – 0.562	$p > 0.562$

* All models are XGBoost classifiers trained on real historical Compustat data (costat = 'I' = distressed).
* Thresholds are industry-calibrated probability cut-offs.
* Higher probability = higher distress risk.

Single Target Assessment

Screening financial health, earnings integrity and valuation of a single company

https://financial-analytics-project-ac820-d1.streamlit.app/

Single Target Assessment

1/3 Safe

American Airlines Group, Inc. (AAL) | FY 2025 | Industrials

Manipulation Risk
Beneish M-Score

Bankruptcy Risk
Logistic Regression

Financial Distress
ISDS-XGBoost (Airline)

Score Summary

Model	Score	Zone / Flag	Direction
Beneish M-Score	**-2.5464**	**Unlikely Manipulator**	Higher (less negative) = riskier
Logistic Regression	**84.28%**	**High Risk**	Probability — lower = safer
ISDS-XGBoost (Airline)	**0.4855**	**Grey Zone**	Distress prob — lower = safer

Detailed Results, Valuation & Model Validation

Beneish M-Score

-2.5464

Unlikely Manipulator

Below -1.78 threshold. No evidence of earnings manipulation.

Logistic Regression

84.28%

High Risk

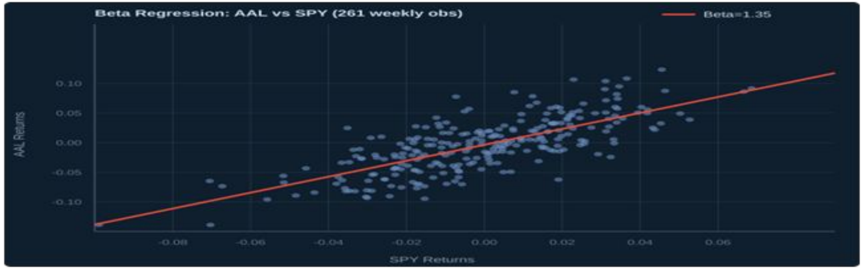
84.3% probability of bankruptcy — high risk, intervention warranted.

ISDS-XGBoost (Airline)

0.4855

Grey Zone

Grey Zone (0.0849–0.9277 band). Elevated risk; monitor quarterly.



CAPM Inputs & Cost of Capital

Risk-Free Rate (Rf)	Market Risk Premium	Expected Growth
4.30%	5.50%	3.00%

Equity Beta	R-Squared	Cost of Equity	Alpha (wkly)
1.346	0.237	11.70%	-0.0037

Intrinsic Value Estimates

Metric	Value	Notes
DCF Fair Value	$10.03	10-yr two-stage DCF, growth fading to terminal
Graham Fair Value	$2.49	Graham: EPS × (8.5 + 2g) × 4.4 / Y
Blended Fair Value	$8.51	Weighted across DCF and Graham methods
Current Price	$12.08	-21.1% vs blended fair value

XGBoost Model Performance (Airline)

Out-of-sample validation on held-out test set

Accuracy	65.8%
Precision	67.1%
Recall	65.8%
F1-Score	66.3%
ROC-AUC	68.2%

Merger Analysis

Quantifying strategic fit and synergy potential between acquirer and target

How the module works

Traditional M&A screening requires juggling multiple tools: distress models, accounting quality checks, and ratio spreadsheets — each in isolation. This module consolidates the full workflow into three sequential steps.

01 Individual Risk Screen

Before evaluating fit, the model checks both companies independently for:

- Earnings manipulation (**Beneish M-Score**)
- Bankruptcy probability (**Logistic Regression**)
- Financial distress (**XGBoost**)

A manipulated target invalidates any synergy estimate.

02 Ratio Comparison

Side-by-side analysis of six key M&A ratios:

- ROA
- Leverage
- Liquidity
- Gross Margin,
- OCF/Assets
- Asset Turnover

Signals indicate relative strength — not absolute health — to identify where complementarity or integration risk lies.

03 Synergy Scorecard

Five dimensions scored independently:

- Revenue Diversification
- Cost/Margin Synergy
- Balance Sheet Fit
- Liquidity Complement
- Size Complement

Verdict: HIGH / MODERATE / LOW SYNERGY.

Comparative Results: 3 M&A Scenarios

Netflix and Warner Bros

Merger Analysis Results

Overall Synergy Assessment

HIGH SYNERGY

Strong synergy potential across multiple dimensions.

Synergy Dimensions

Dimension	Synergy Level	Explanation
Revenue Diversification	High	Significantly different growth profiles create diversification.
Cost / Margin Synergy	High	Large margin gap suggests cost restructuring opportunity.
Financial / Balance Sheet	No	Both entities carry significant leverage — limited balance-sheet synergy.
Liquidity Complement	Low	Adequate combined liquidity.
Size Complement	High	Appropriate size difference for bolt-on integration.

Key Financial Ratios Comparison

Side-by-side comparison of the most relevant M&A screening ratios. Gap = Acquirer minus Target. Signal indicates strategic fit.

Ratio	Category	Netflix, Inc.	Warner Bros. Di	Signal
ROA (Net Income / Assets)	Profitability	19.8%	0.7%	✅ Acquirer stronger
Leverage (Liabilities / Assets)	Solvency	52.1%	62.9%	✅ Acquirer less leveraged
Liquidity (Current Ratio)	Liquidity	1.19x	1.06x	✅ Acquirer stronger
Gross Margin	Profitability	48.5%	44.0%	➡️ Similar
OCF / Assets	Cash Generation	18.3%	4.3%	✅ Acquirer stronger
Asset Turnover (Rev / Assets)	Efficiency	0.81x	0.37x	✅ Acquirer stronger

Union Pacific and Norfolk Southern

Merger Analysis Results

Overall Synergy Assessment

MODERATE SYNERGY

Some synergy opportunities exist but integration risk is present.

Synergy Dimensions

Dimension	Synergy Level	Explanation
Revenue Diversification	No	Similar growth profiles — limited diversification benefit.
Cost / Margin Synergy	Low	Moderate margin differential.
Financial / Balance Sheet	No	Both entities carry significant leverage — limited balance-sheet synergy.
Liquidity Complement	No	Both entities have tight liquidity — integration funding risk.
Size Complement	High	Appropriate size difference for bolt-on integration.

Key Financial Ratios Comparison

Side-by-side comparison of the most relevant M&A screening ratios. Gap = Acquirer minus Target. Signal indicates strategic fit.

Ratio	Category	Union Pacific C	Norfolk Souther	Signal
ROA (Net Income / Assets)	Profitability	10.2%	6.4%	➡️ Similar
Leverage (Liabilities / Assets)	Solvency	73.5%	65.6%	⚠️ Acquirer more leveraged
Liquidity (Current Ratio)	Liquidity	0.91x	0.85x	✅ Acquirer stronger
Gross Margin	Profitability	45.8%	36.3%	✅ Acquirer stronger
OCF / Assets	Cash Generation	13.3%	9.6%	➡️ Similar
Asset Turnover (Rev / Assets)	Efficiency	0.35x	0.27x	✅ Acquirer stronger

United Airlines and American Airlines

Merger Analysis Results

Overall Synergy Assessment

LOW SYNERGY

Limited synergy potential — proceed with caution.

Synergy Dimensions

Dimension	Synergy Level	Explanation
Revenue Diversification	No	Similar growth profiles — limited diversification benefit.
Cost / Margin Synergy	Low	Moderate margin differential.
Financial / Balance Sheet	No	Both entities carry significant leverage — limited balance-sheet synergy.
Liquidity Complement	No	Both entities have tight liquidity — integration funding risk.
Size Complement	Low	Merger of equals — complex integration but transformative potential.

Key Financial Ratios Comparison

Side-by-side comparison of the most relevant M&A screening ratios. Gap = Acquirer minus Target. Signal indicates strategic fit.

Ratio	Category	United Airlines	American Airlin	Signal
ROA (Net Income / Assets)	Profitability	4.4%	0.2%	➡️ Similar
Leverage (Liabilities / Assets)	Solvency	80.0%	106.0%	✅ Acquirer less leveraged
Liquidity (Current Ratio)	Liquidity	0.65x	0.50x	✅ Acquirer stronger
Gross Margin	Profitability	28.8%	19.2%	✅ Acquirer stronger
OCF / Assets	Cash Generation	11.0%	3.1%	✅ Acquirer stronger
Asset Turnover (Rev / Assets)	Efficiency	0.77x	0.88x	⚠️ Target stronger

Synergy Dimensions: Comparative Results

How the model scores each dimension across the 3 M&A scenarios

	HIGH SYNERGY	MODERATE SYNERGY	LOW SYNERGY
Synergy Dimension	**Netflix · Warner Bros Discovery**	**Union Pacific · Norfolk Southern**	**United Airlines · American Airlines**
Revenue Diversification	HIGH Very different growth profiles → real diversification value.	NO Same rail sector, similar growth → no diversification.	NO Same routes & business model → zero diversification.
Cost / Margin Synergy	HIGH Large margin gap (48.5% vs 44%) → major restructuring opportunity.	LOW Moderate gap (45.8% vs 36.3%) → some operational upside.	LOW Gap exists (28.8% vs 19.2%) but insufficient to drive synergy.
Financial / Balance Sheet	NO Both leveraged (52% vs 63%) → limited balance-sheet synergy.	NO UNP more leveraged than target (73.5% vs 65.6%) → added risk.	NO AAL at 106% debt/assets — critically over-leveraged. Worst case.
Liquidity Complement	LOW Adequate combined liquidity (1.19x vs 1.06x) → manageable.	NO Both tight → integration funding risk flagged by the model.	NO Both very tight (0.65x vs 0.50x) → high integration funding risk.
Size Complement	HIGH Appropriate size difference → clean bolt-on integration.	HIGH ~$130B acquires ~$55B → ideal bolt-on size ratio.	LOW Merger of equals → complex integration, no scale advantage.

Textual Sentiment Analysis

Extracting risk signals from corporate disclosures using NLP

Textual Sentiment Analysis NVDA VS AAPL

NVDA | 2025

Fetch & Analyse

Fetched 200,000 characters of 10-K text.

Sentiment Results

NVDA (2025)

Sentiment Prediction

Mild Positive Sentiment

Confidence: 62.4%

The filing language is moderately positive. Some caution is present but overall tone suggests stability.

Readability Indexes — NVDA (2025)

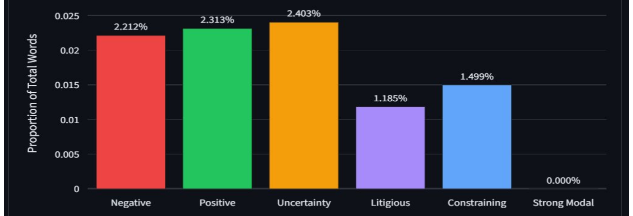
Flesch-Kincaid Grade Level	Gunning Fog Index	Automated Readability Index (A...
17.9	21.9	18.8
↑ Grade 18	↑ Grade 22	↑ Grade 19
Post-graduate level. Dense but normal for 10-K filings.	Extremely difficult. Dense legal/technical prose — may signal obfuscation.	Post-graduate / professional level. Extremely dense prose.

Index	Component	Value
Flesch-Kincaid Grade Level	Avg Sentence Length	28.9
Flesch-Kincaid Grade Level	Avg Syllables Per Word	1.88
Gunning Fog Index	Avg Sentence Length	28.9
Gunning Fog Index	Pct Complex Words	25.8
Automated Readability Index (ARI)	Avg Chars Per Word	5.48
Automated Readability Index (ARI)	Avg Sentence Length	28

L&M Dictionary Features — NVDA (2025)

Negative	Positive	Uncertainty	Litigious	Constraining	Strong Modal
2.21%	2.31%	2.40%	1.18%	1.50%	0.00%
↑ 661 words	↑ 691 words	↑ 718 words	↑ 354 words	↑ 448 words	↑ 0 words

L&M Word Category Proportions



Top 5 Most Frequent Words by Category

The most repeated words from each L&M dictionary category found in the filing.

Negative	Positive	Uncertainty	Litigious	Constraining	Strong Modal
harm 40x	certain 85x	may 303x	laws 44x	requirements 64x	No words found
risk 30x	solutions 38x	could 176x	regulations 33x	restrictions 50x	
adversely 28x	support 34x	believe 18x	regulatory 28x	subject 47x	
adverse 27x	competitive 28x	potential 17x	legal 23x	limited 37x	
unable 23x	able 22x	volatility 16x	liability 17x	compliance 28x	

Total words analysed: 29,880 | Net Sentiment (Pos-Neg): 0.0010

Text Preview — NVDA (2025)

nvda- Table of Contents UNITED STATES SECURITIES AND EXCHANGE COMMISSION Washington, D.C. 20549

______________ FORM 10-K ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934 For the fiscal year ended January 26 , 2025 OR TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934 Commission file number: 0-23985 NVIDIA CORP ORATION (Exact name of registrant as specifi... in its charter) Delaware 94-3177549 (State or other jurisdiction of (I.R.S. Employer inco...rati...

AAPL | 2025

Fetch & Analyse

Fetched 200,000 characters of 10-K text.

Sentiment Results

AAPL (2025)

Sentiment Prediction

Mild Negative Sentiment

Confidence: 33.8%

The filing language leans negative. Elevated uncertainty, risk, or litigation language detected — may signal emerging distress.

Readability Indexes — AAPL (2025)

Flesch-Kincaid Grade Level	Gunning Fog Index	Automated Readability Index (ARI)
17.6	22.5	18.0
↑ Grade 18	↑ Grade 22	↑ Grade 18
Post-graduate level. Dense but normal for 10-K filings.	Extremely difficult. Dense legal/technical prose — may signal obfuscation.	Post-graduate / professional level. Extremely dense prose.

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Flesch-Kincaid Grade Level	Avg Syllables Per Word	1.93
Gunning Fog Index	Avg Sentence Length	26.7
Gunning Fog Index	Pct Complex Words	29.5
Automated Readability Index (ARI)	Avg Chars Per Word	5.

M&A Risk Analyzer

A Quantitative framework for screening acquisition targets

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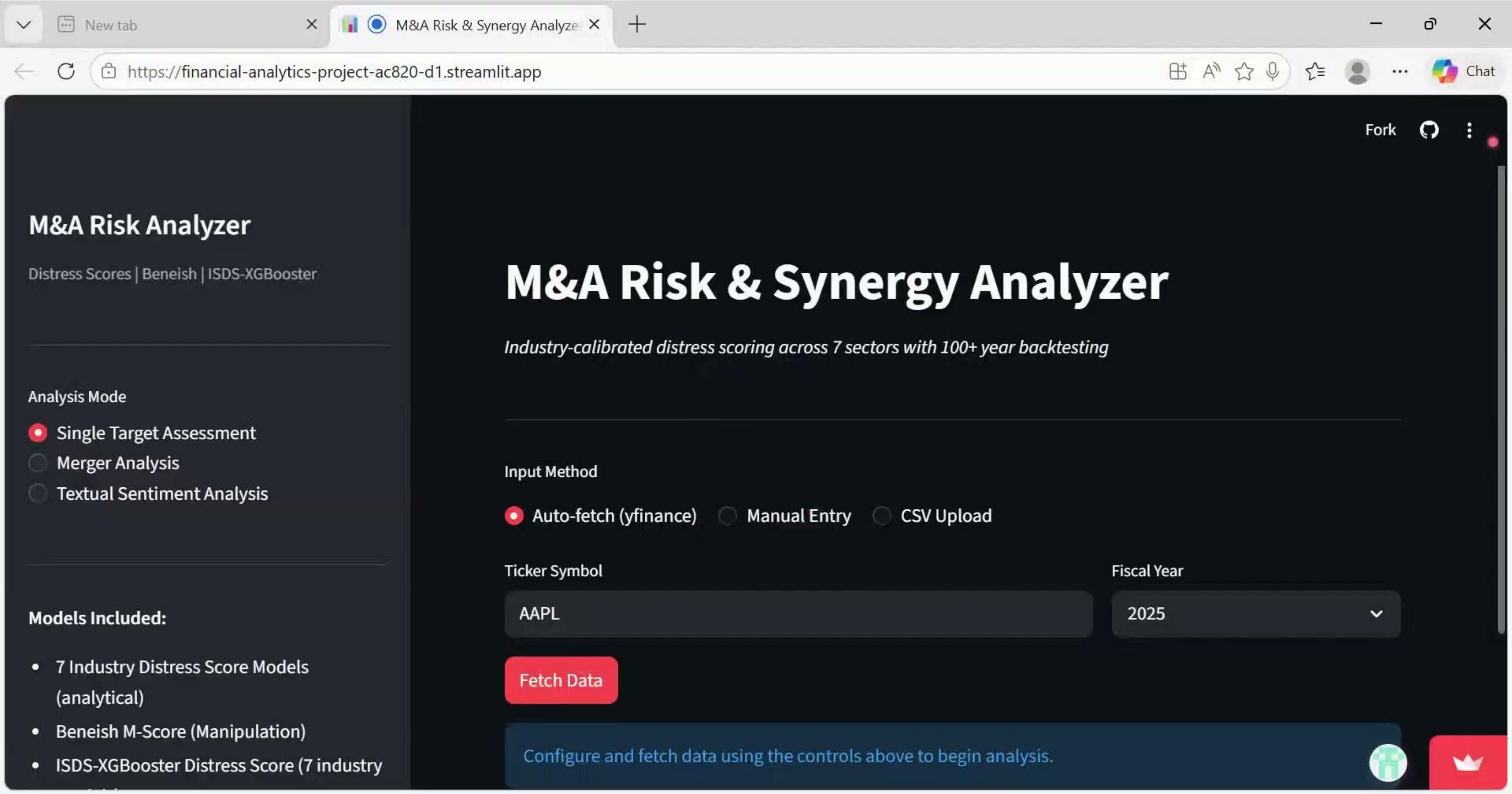
How the Synergy Scorecard is calculated

All inputs from yfinance (annual financials). Five dimensions scored independently. Verdict: ≥3 HIGH → HIGH SYNERGY | ≥1 HIGH or ≥3 LOW → MODERATE | else → LOW

Dimension	Metric used	HIGH if...	LOW if...	NO if... / Example
Revenue Diversification	Δ Revenue growth rate (revenue / revenue_prev − 1)	**Growth gap > 15pp**	**Growth gap 5–15pp**	**Growth gap < 5pp** *e.g. NFLX 14.1% vs WBD 3% → gap 11pp → LOW*
Cost / Margin Synergy	EBIT margin (EBIT / Revenue)	**Margin gap > 10pp**	**Margin gap 3–10pp**	**Margin gap < 3pp** *e.g. NFLX 22% vs WBD 3% → gap 19pp → HIGH*
Financial / Balance Sheet	Leverage (Total Liabilities / Total Assets)	**Acquirer lev < 0.50 AND target lev > 0.60**	**Average leverage < 0.55**	**Average leverage ≥ 0.55** *e.g. UAL 80% + AAL 106% → avg 93% → NO*
Liquidity Complement	Current Ratio (Current Assets / Current Liabilities)	**Either company current ratio > 1.5x**	**Both > 1.0x**	**Both ≤ 1.0x** *e.g. NFLX 1.19x, WBD 1.06x → both > 1 → LOW*
Size Complement	size_ratio = min(Total Assets A, Total Assets B) / max(Total Assets A, Total Assets B)	**Ratio between 0.20 and 0.80**	**Ratio ≥ 0.80 (merger of equals)**	**Ratio < 0.20 (too large a gap)** *e.g. UNP $80B / NSC $34B → ratio 0.43 → HIGH*


New tab
M&A Risk & Synergy Analyze
https://financial-analytics-project-ac820-d1.streamlit.app
Chat
Fork
M&A Risk Analyzer
Distress Scores | Beneish | ISDS-XGBooster
Analysis Mode
Single Target Assessment
Merger Analysis
Textual Sentiment Analysis
Models Included:
7 Industry Distress Score Models (analytical)
Beneish M-Score (Manipulation)
ISDS-XGBooster Distress Score (7 industry
M&A Risk & Synergy Analyzer
Industry-calibrated distress scoring across 7 sectors with 100+ year backtesting
Input Method
Auto-fetch (yfinance)
Manual Entry
CSV Upload
Ticker Symbol
AAPL
Fiscal Year
2025
Fetch Data
Configure and fetch data using the controls above to begin analysis.